

[11, Assumption 4.1] imposes additional conditions, such as $f \in \mathcal{C}^2(\mathbb{R}^d)$ and the boundedness of $\|\nabla^2 f\|_\infty$. In the framework of [22, Definition 8], some additional local growth conditions on f around the global minimizer are required. For our rescaled CBO we do not require these additional assumptions for the convergence proof.

- **Multiple–minimizer case:** Although our rescaled CBO is not specifically designed to address the multiple minimizer problem, we can still establish certain asymptotic properties in cases where the objective function $f(\cdot)$ has multiple minimizers, whether they are discrete points or compact subsets (see Theorem 3.9). Let us mention here that specially designed CBO models aimed at addressing multiple-minimizer problems have been introduced in [9, 24].

Our results rely on two main ingredients: uniform estimates on the moments of the measure, and on a Harnack inequality on the invariant elliptic CBO equation. Indeed we need Harnack’s inequality in order to ensure that the measure has a non–null mass around minimizers, which will then allow us to prove Laplace’s principle. A summary of our findings is in **section 4** at the end of the manuscript.

2. USEFUL ESTIMATES

We start by recalling some notation. When a measure μ has a density ϱ with respect to Lebesgue measure that we denote by dx , then μ is absolutely continuous with respect to dx , we write $\mu \ll dx$ and $\varrho = \frac{d\mu}{dx}$ is Radon–Nikodym derivative of μ with respect to dx . Unless a confusion arises, we shall use the same notation for a measure and for its density (when it exists). We shall also denote by $W^{k,p}(\Omega)$, $p \geq 1, k \geq 0$ the standard Sobolev space of functions whose generalized derivatives up to order k are in $L^p(\Omega)$, and we denote by $W_{loc}^{k,p}(\Omega)$ the class of functions f such that $\chi f \in W^{k,p}(\Omega)$ for each $\chi \in \mathcal{C}_c^\infty(\Omega)$ the class of infinitely differentiable functions with compact support in Ω .

In what follows, $\|\cdot\|$ denotes the Frobenius norm of a matrix and $|\cdot|$ is the standard Euclidean norm in $\mathbb{R}^d$; $\mathcal{P}(\mathbb{R}^d)$ denotes the space of probability measures on $\mathbb{R}^d$, and $\mathcal{P}_p(\mathbb{R}^d)$ with $p \geq 1$ contains all $\mu \in \mathcal{P}(\mathbb{R}^d)$ such that $\mu(|\cdot|^p) := \int_{\mathbb{R}^d} |x|^p \mu(dx) < \infty$; it is equipped with p –Wasserstein distance $W_p(\cdot, \cdot)$.

Our main assumption concerns the function $f(\cdot)$ to be minimized, and can be stated as follows.

Assumption 2.1. *We assume the following properties for the objective function.*

- (1) $f : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is bounded from below by $\underline{f} := \min f > 0$, and there exists some constant $L_f > 0$ such that

$$|f(x) - f(y)| \leq L_f(1 + |x| + |y|)|x - y|, \quad \forall x, y \in \mathbb{R}^d.$$

- (2) There exist constants $c_\ell, c_u > 0$ and $M > 0$ such that

$$f - \underline{f} \leq c_u(|x|^2 + 1), \quad \forall x \in \mathbb{R}^d \quad \text{and} \quad c_\ell|x|^2 \leq f - \underline{f} \quad \text{for } |x| \geq M.$$